

UNIT 7

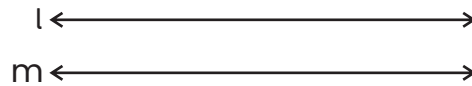
FUNDAMENTALS OF GEOMETRY



PARALLEL LINES

Two lines are said to be parallel if

- (i) They are coplanar
- (ii) They do not intersect.



Parallel is denoted by ' $\parallel$ '. $l \parallel m$ is read as 'l is parallel to m'

Properties of parallel lines:

Property 1: Two lines which are parallel to the same given line are parallel to each other. Look at the figures given below:



In the figures given above $l \parallel n$ and $m \parallel n$, therefore $l \parallel m$. This is an important property of parallel lines.

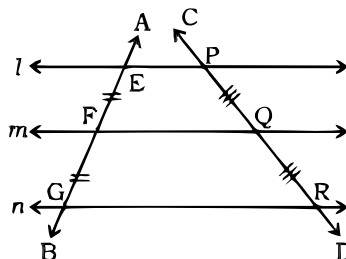
Property 2: If three parallel lines are intersected by two transversals in such a way that the two intercepts on one transversal are equal to each other, then the two intercepts on the second transversal are also equal.

Look at the figure given on the right.

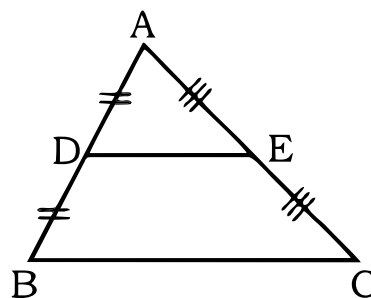
There are three lines l , m and n parallel to one another.

If $m\overline{EF} = m\overline{FG}$

Then $m\overline{PQ} = m\overline{QR}$



Property 3. A line through the midpoint of two sides of a triangle parallel to the third side.



In $\triangle ABC$, D is the midpoint of $\overline{AB}$ and $\overline{DE} \parallel \overline{BC}$

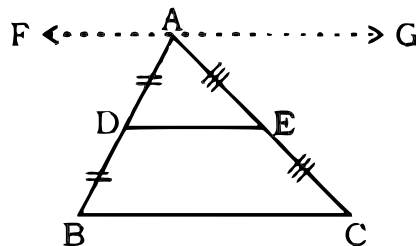
That is: $m\overline{AD} = m\overline{DB}$

Then by Property 2, point E bisects $\overline{AC}$ (the third side)

That is: $m\overline{AE} = m\overline{EC}$

Now look at figure (ii)

$\overline{DE} \parallel \overline{BC}$



Through point A, line FG has been drawn parallel to line DE.

When a transversal intersects two or more parallel lines

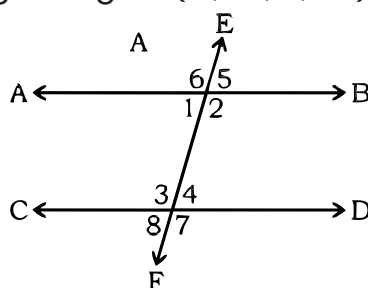
(i) Vertically Opposite Angles are equal

(ii) Corresponding Angles are equal

(iii) Alternate Interior Angles are equal

(iv) Interior Angles on the same side of the transversal are supplementary

In the figure given here, two parallel line AB and CD are intersected by the transversal EF as result eight angles ($\angle 1, \angle 2, \dots, \angle 8$) are formed.



Out of these Angles, $\angle 1, \angle 2, \angle 3$ and $\angle 4$ are Interior Angles, whereas $\angle 5, \angle 6, \angle 7$ and $\angle 8$ are Exterior Angles. ($\angle 1, \angle 5$); ($\angle 2, \angle 6$); ($\angle 3, \angle 7$) and ($\angle 4, \angle 8$) are four pairs of Vertically Opposite Angles.

Pairs of Corresponding Angles are:

(i) $\angle 4, \angle 5$

(ii) $\angle 3, \angle 6$

(iii) $\angle 1, \angle 8$

(iv) $\angle 2, \angle 7$

Two pairs of Alternate Interior Angles are:

(i) $\angle 1$ and $\angle 4$

(ii) $\angle 2$ and $\angle 3$

Two pairs of Interior Angles on the same side of the transversal are:

(i) $\angle 1, \angle 3$

(ii) $\angle 2, \angle 4$

Example 1. If two parallel lines AB and CD are intersected by a transversal EF at points G and H, then write the following pair of angles.

- (i) Vertically Opposite angles (ii) Corresponding angles
 (iii) Alternate Interior angles
 (iv) Interior Angles on the same side of the transversal

Solution: (i) Four pairs of Vertically Opposite Angles are:

- (a) $\angle AGH, \angle EGB$ (b) $\angle AGE, \angle BGH$
 (c) $\angle CHF, \angle GHD$ (d) $\angle GHC, \angle DHF$

(ii) Four pairs of Corresponding Angles are:

- (a) $\angle AGH, \angle CHF$ (b) $\angle AGE, \angle GHC$
 (c) $\angle EGB, \angle GHD$ (d) $\angle BGH, \angle DHF$

(iii) Two pairs of Alternate Interior Angles are:

- (a) $\angle AGH, \angle GHD$ (b) $\angle BGH, \angle GHC$

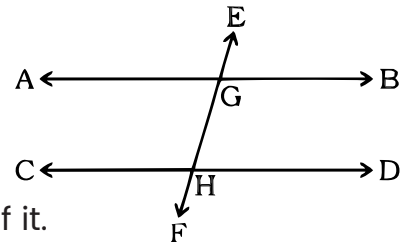
(iv) Two pairs of Interior Angles on the same side of transversal are:

- (a) $\angle AGH, \angle GHC$ (b) $\angle BGH, \angle GHD$

Relations between the pairs of angles when a transversal intersects two parallel lines

Example 1. In the figure on the right, identify the angle which is:

- (i) Congruent to $\angle GHD$ being corresponding angle to it.
 (ii) Congruent to $\angle GHD$ being an alternate angle of it.
 (iii) Supplement of $\angle GHD$ being
 (a) an interior angle on the same side of the transversal
 (b) an interior angle adjacent to it
 (c) an exterior angle adjacent to it
 (iv) Congruent to $\angle GHD$ being its vertically opposite angle



Solution:

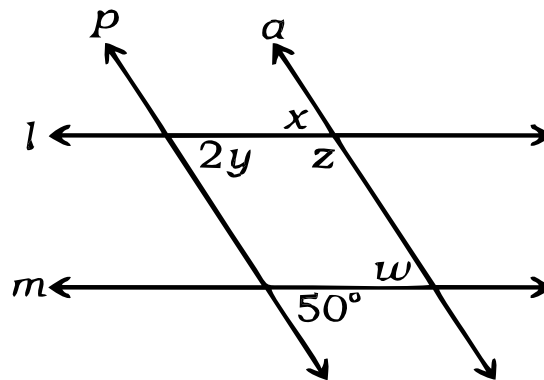
- (i) $\angle EGB \cong \angle GHD$ (Corresponding angles)
 (ii) $\angle AGH \cong \angle GHD$ (Alternate angles)
 (iii) (a) $\angle BGH$ is supplement of angle GHD (Interior angle on the same side of the transversal)
 (b) $\angle GHC$ is supplement of angle GHD (Interior angle adjacent to it)
 (c) $\angle DHF$ is supplement of $\angle GHD$ (Exterior angle adjacent to it)
 (iv) $\angle CHF \cong \angle GHD$ (Vertically opposite angles)

Example 2. In the figure of Example 2, find the remaining angles when $m\angle GHD = 70^\circ$

Solution:

- (1) $\angle AGH \cong \angle GHD$ (Alternate angles) $\therefore m\angle AGH = 70^\circ$
- (2) $\angle EGB \cong \angle GHD$ (Corresponding angles) $\therefore m\angle EGB = 70^\circ$
- (3) $m\angle BGH + m\angle GHD = 180^\circ$ (Interior angle on the same side of transversal)
 $\Rightarrow m\angle BGH = 180^\circ - 70^\circ = 110^\circ$
- (4) $\angle AGE \cong \angle BGH$ (Vertically Opposite angles) $\therefore m\angle AGE = 110^\circ$
- (5) $\angle GHC \cong \angle BGH$ (Alternate angles) $\therefore m\angle GHC = 110^\circ$
- (6) $\angle DHF \cong \angle GHC$ (Vertically Opposite angles) $\therefore m\angle DHF = 110^\circ$
- (7) $\angle CHF \cong \angle GHD$ (Vertically Opposite angles) $\therefore m\angle CHF = 70^\circ$

Example 3. Find the values of $\angle w$, $\angle x$, $\angle y$, and $\angle z$.

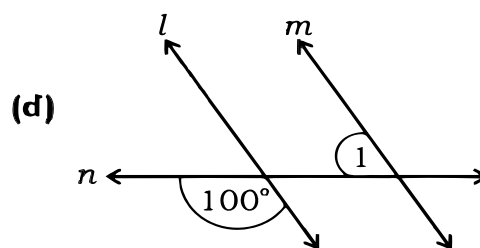
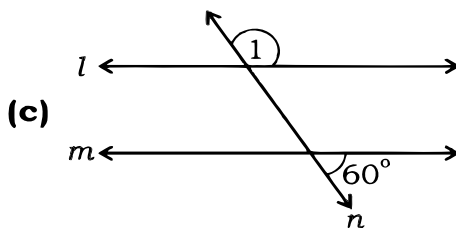
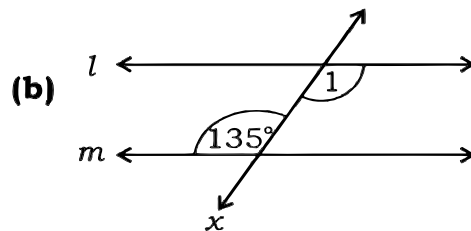
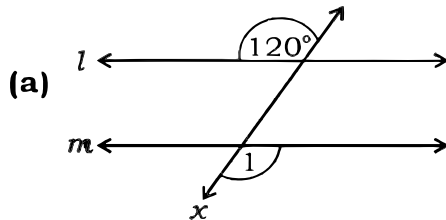


Solution:

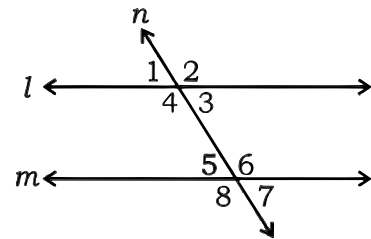
- (1) w is an alternate angle of 50°
 $\therefore m\angle w = 50^\circ$
- (2) x and w are corresponding angles
 $\therefore m\angle x = m\angle w \Rightarrow m\angle x = 50^\circ$
- (3) $\angle 2y$ is an alternate angle of x
 $\therefore m\angle 2y = m\angle x \Rightarrow m\angle 2y = 50^\circ$ or $m\angle y = 50^\circ/2 = 25^\circ$
- (4) $\angle z$ is supplement of $\angle x \Rightarrow m\angle z = 180^\circ - 50^\circ = 130^\circ$

EXERCISE 7.1

Q. Find the measure of $\angle 1$ in each of the figures below when measure of one angle is given.



2. In the adjacent figure $l \parallel m$. Find the measure of remaining angles when $m\angle 3 = 65^\circ$



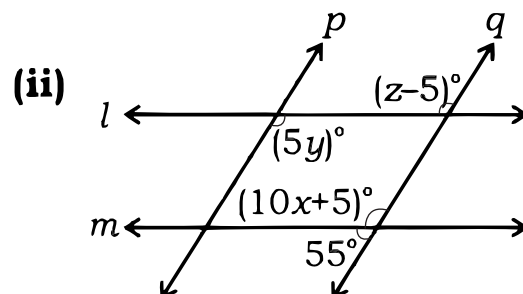
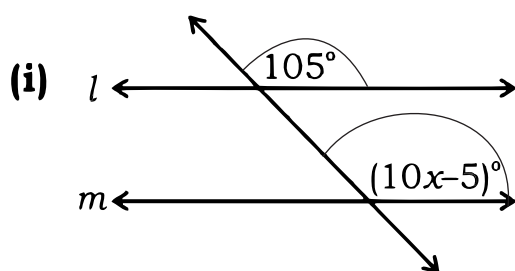
3. $AB \parallel CD$ and EF intersects them at points P and Q respectively, $m\angle PQD = 50^\circ$. Draw the figure and find the measure of remaining angles.

4. $AB \parallel CD$ and EF intersects them at points G and H respectively. Identify the angle which is:

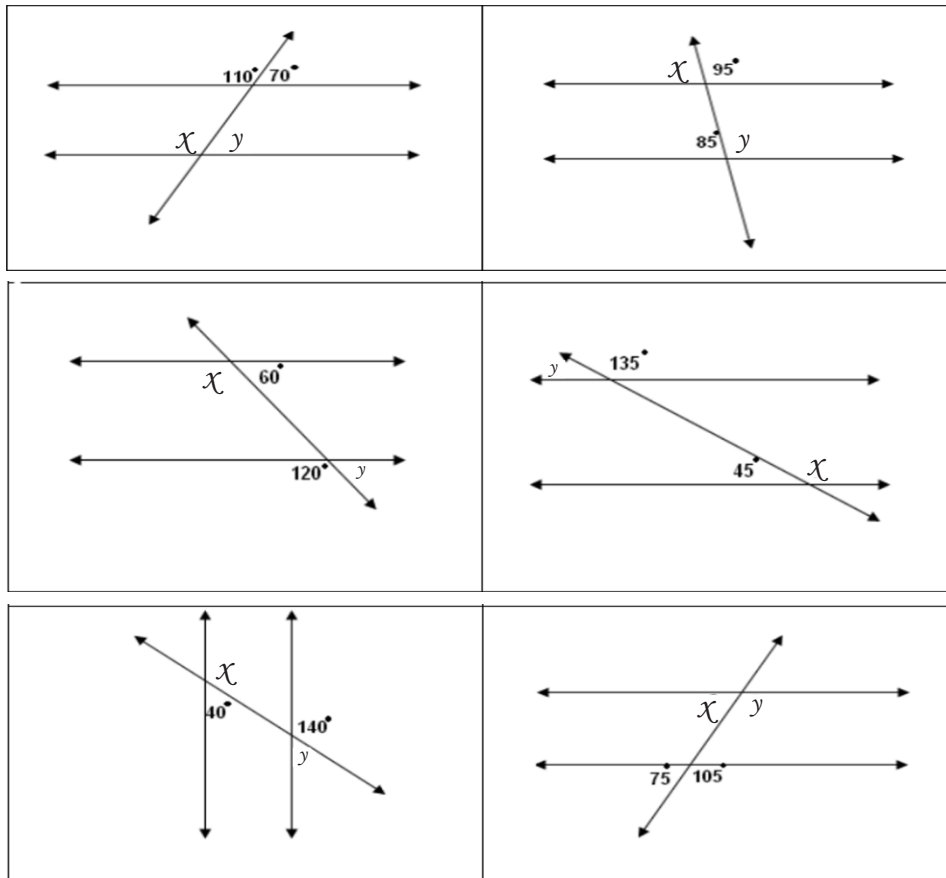
- (i) Congruent to $\angle GHC$ as a corresponding angle
- (ii) Congruent to $\angle GHC$ as an alternate angle
- (iii) Congruent to $\angle GHC$ as a vertically opposite angle
- (iv) Supplement of $\angle GHC$:

- (a) as an interior angle on the same side of the transversal
- (b) as an interior angle adjacent to it
- (c) as an exterior angle adjacent to it.

5. Solve for x , y , or z and find the angle in each case when $l \parallel m$ and $p \parallel q$.

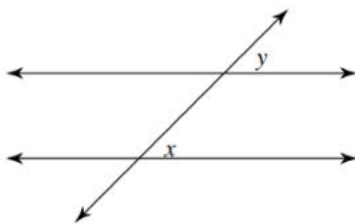


Q. Find the missing angles:

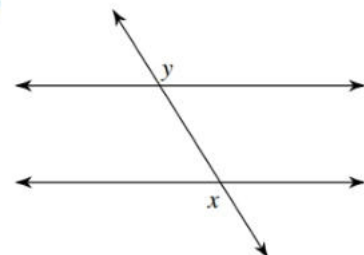


Q. Identify each pair of angles as corresponding, alternate interior, alternate exterior, or consecutive interior.

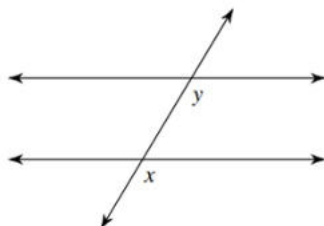
1)



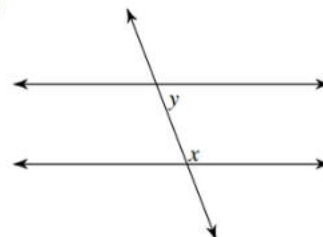
2)



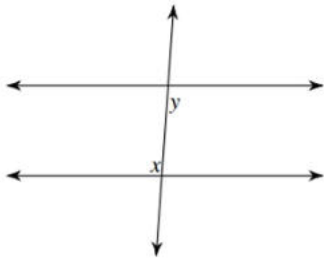
3)



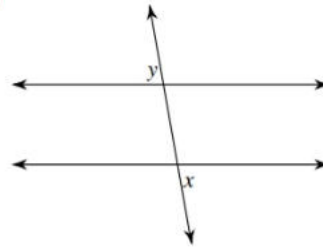
4)



5)

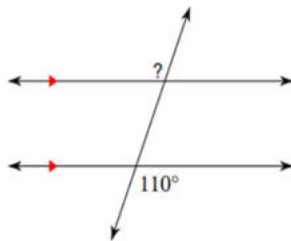


6)

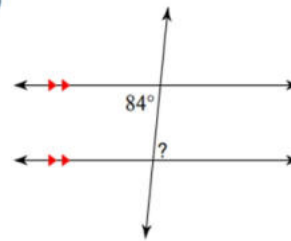


Q. Find the measure of each angle indicated.

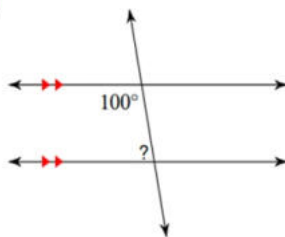
1)



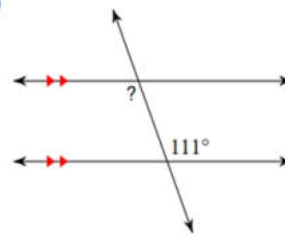
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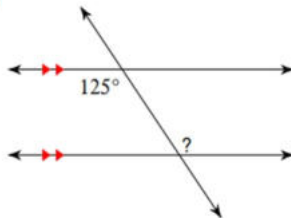
3)



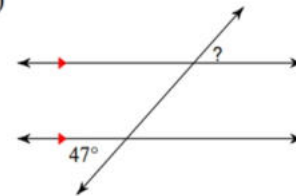
4)



5)

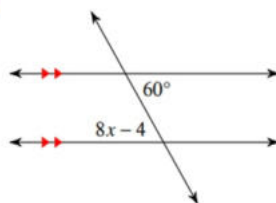


6)

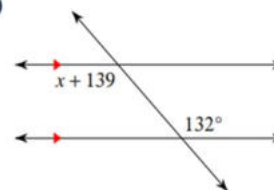


Q. Solve for x.

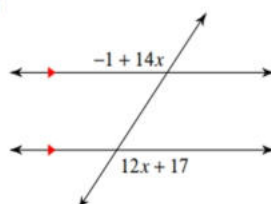
1)



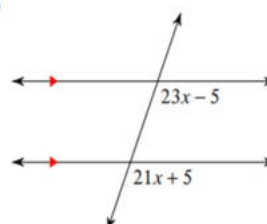
2)



3)



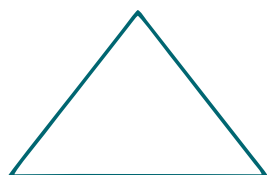
4)



POLYGONS

Definition of a Polygon

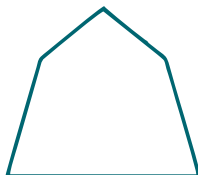
A plane closed figure bounded by three or more sides is called a **Polygon**. Figures of some important polygons along with their names are given here.



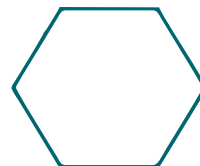
Triangle
(Tri means three)



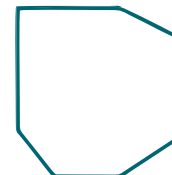
Quadrilateral
(Quad means four)



Pentagon
(Penta means five)



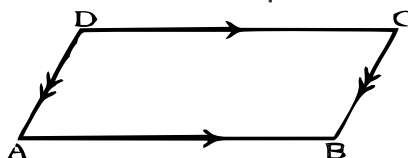
Hexagon
(Hexa means six)



Heptagon
(Hepta means seven)

Properties of a parallelogram

Parallelogram is a quadrilateral whose opposite sides are parallel. Opposite sides of a parallelogram are equal. Diagonals of a parallelogram bisect each other. For example, the figure shown here is a quadrilateral.

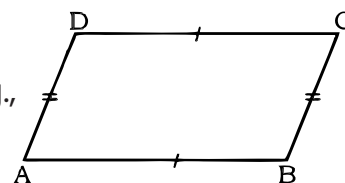


Quad. ABCD is a parallelogram. Symbolically a parallelogram is denoted as $\parallel$.

A parallelogram has the following properties:

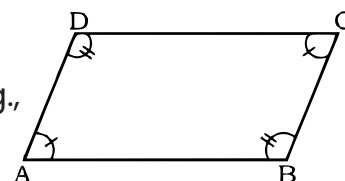
1. Opposite sides are equal.

Opposite sides of a parallelogram are congruent, e.g.,
In parallelogram ABCD,
 $\overline{AB} \cong \overline{CD}$ and $\overline{AD} \cong \overline{BC}$



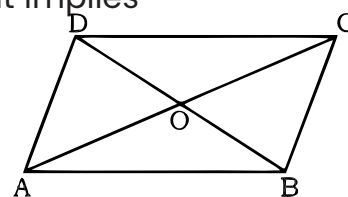
2. Opposite angles are congruent.

Opposite angles of parallelogram are congruent, e.g.,
in parallelogram ABCD $\angle A \cong \angle C$ and $\angle B \cong \angle D$



3. Diagonals bisect each other.

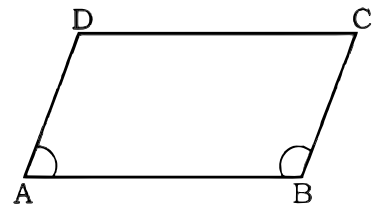
Diagonals of parallelogram bisect one another. For example,
In parallelogram ABCD, diagonal $\overline{AC}$ intersects diagonal $\overline{BD}$ at O in such a way that O is the midpoint of both $\overline{AC}$ and $\overline{BD}$. It implies that $\overline{AC}$ and $\overline{BD}$ bisect one another, i.e., $\overline{AO} = \overline{CO}$ and $\overline{BO} = \overline{DO}$ and $m\overline{AO} = \frac{1}{2} m\overline{AC} = m\overline{CO}$ and $m\overline{BO} = \frac{1}{2} m\overline{BD} = m\overline{DO}$



4. Angles at the end-points of each side are supplementary

Each pair of consecutive angles (i.e., angles at the end-point of each side)

of a parallelogram are supplementary. e.g., In parallelogram ABCD, $m\angle A + m\angle B = 180^\circ$. Similarly,
 $m\angle B + m\angle C = m\angle C + m\angle D = m\angle D + m\angle A = 180^\circ$



Example 1. In parallelogram PQRS, $m\overline{PQ} = 5$ cm, $m\overline{QR} = 3$ cm and $m\angle P = 70^\circ$, find the length of other two sides and measure of remaining angles.

Solution:

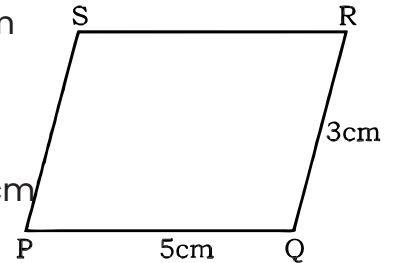
$\overline{RS} \cong \overline{PQ}$ (opposite sides of a parallelogram) and $m\overline{PQ} = 5$ cm
 $\therefore m\overline{RS} = 5$ cm

$\overline{PS} \cong \overline{QR}$ (opposite sides of a parallelogram) and $m\overline{QR} = 3$ cm
 $\therefore m\overline{PS} = 3$ cm

$\angle R \cong \angle P$ (opposite angles of a parallelogram) and $m\angle P = 70^\circ$
 $\therefore m\angle R = 70^\circ$

Further $m\angle P + m\angle Q = 180^\circ$ (Interior angles on the same side of transversal)
 $\Rightarrow m\angle Q = 180^\circ - 70^\circ$ ($\because m\angle P = 70^\circ$)
 $\Rightarrow m\angle Q = 110^\circ$

Also, $m\angle S = m\angle Q$ (opposite angles of a parallelogram) and $m\angle Q = 110^\circ$
 $\therefore m\angle S = 110^\circ$

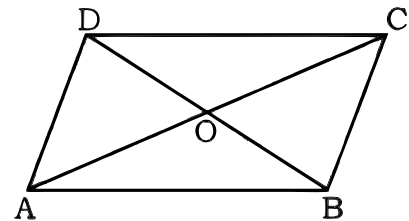


Example 2. In parallelogram ABCD, $m\overline{BD} = 4$ cm and $m\overline{AC}$ is $\frac{3}{2}$ times of $m\overline{BD}$. Both diagonals intersect at O. Find the measure of $\overline{OC}$ and $\overline{OD}$.

Solution:

$m\overline{OD} = \frac{1}{2} m\overline{BD} = \frac{1}{2} (4 \text{ cm}) = 2 \text{ cm}$
 $\therefore m\overline{OD} = 2 \text{ cm}$

Further $m\overline{AC} = \frac{3}{2}$ of $m\overline{BD} = \frac{3}{2} \times 4 = 6 \text{ cm}$
 $\therefore m\overline{OC} = \frac{1}{2} m\overline{AC} = \frac{1}{2} (6) = 3 \text{ cm}$



Regular Polygon:

A polygon in which all the sides are congruent (equal in measure) to one another is called a Regular Polygon. In a regular polygon all interior angles are congruent to one another.

Regular Pentagon:

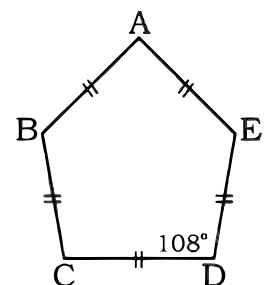
A pentagon (i.e., five sided closed figure) in which all sides are congruent to each other is called a Regular Pentagon.

Sum of measures all the angles of a polygon is

$$S = (2n - 4) 90^\circ$$

$$= (2(5) - 4) 90 = (10 - 4) 90 = 6 \times 90 = 6 \times 90 = 540^\circ$$

$$\therefore \text{Measure of each angle of a Regular Pentagon} = 540/5 = 108^\circ$$



Regular Hexagon:

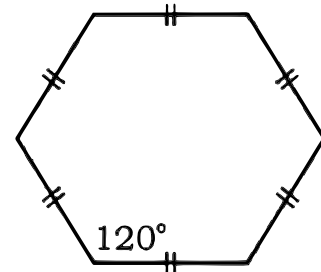
A hexagon (six-sided closed figure) in which all sides are congruent to each other is called a Regular Hexagon.

Sum of measure of all the angles of a Hexagon is:

$$S = (2n - 4) 90^\circ = (2(6) - 4) 90$$

$$= (12 - 4) 90 = 8 \times 90 = 720^\circ$$

Each angle of a Regular Hexagon measures = $720^\circ / 6 = 120^\circ$



Regular Octagon:

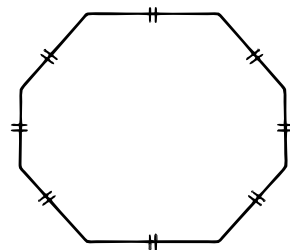
An octagon (eight sided closed figure) all of whose sides are congruent to one another is called a Regular Octagon.

Sum of measures of all the angles of a Regular Octagon is:

$$S = (2n - 4) 90^\circ$$

$$S = (2(8) - 4) 90 = (16 - 4) 90 = 12 \times 90 = 1080^\circ$$

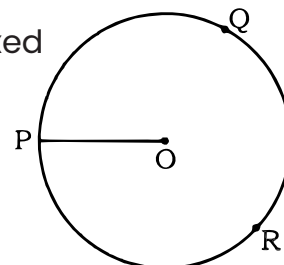
.. Measure of each angle of a regular octagon = $1080 \div 8 = 135^\circ$



CIRCLE

A circle is a set of points in a plane equidistant from a fixed point called its centre.

Here PQR is a circle whose centre is O. Line OP is the radial segment and $m \overline{OP}$ is the radius of the circle.



Demonstrate a point lying in the interior and exterior of a circle.

(1) Set of points of the circle: It consists of all these points whose distances from the centre are equal to its radius. Points P, Q, R belong to this set. This set of points is called the boundary of the circle.

(2) Set of points lying inside the circle: It consists of all the parts whose distances from the centre are less than its radius. This set of points is called the Interior of the Circle.

(3) Set of points lying outside the circle: It consists of all the points whose distances from the centre are greater than its radius. This set of points is called the Exterior of the Circle.

Terms related to a circle

Arc: Any portion or part of the boundary of a circle is called an arc of the circle. Arc may be minor or major. An arc is said to be a Minor Arc if it is less than a Semi-circle (Half Circle). An arc is said to be a Major Arc if it is greater than a semi-circle.

Chord: A line segment joining any two points of a circle is called a chord of a circle. A chord passing through the centre is called a Diameter.

Measure of a Diameter = 2 radius

Secant: A line which intersects the circle at any two of its point is called a secant. In figure (i) line m is a secant.

Tangent: A line (line segment or ray) is called a tangent to the circle if it intersects the circle at one and only one point, i.e., it touches the circle at one point only.

In figure (i) $\overleftrightarrow{AB}$, $\overleftrightarrow{QR}$ and $\overline{CD}$ are tangents to the circle at points P, Q and T.

Sector: A portion of circular region bounded by two radial segments and an arc is called a sector. In figure (ii), the shaded portion is a sector of a circle whose centre is O. This sector is bounded by radial segments OA and OB and minor arc AB.

Concyclic Points: If four or more points are such that a circle passes through all the four (or more) points, these points are called concyclic points. In figure (iii) points A, B, C and D are concyclic points. The vertices of a quadrilateral are concyclic only when their opposite angles are supplementary.

Concentric circles: Two or more circles having the same centre and different radii are called concentric circles.

In figure (v) $\odot ABC$, $\odot PQR$ and $\odot XYZ$ are concentric circles.

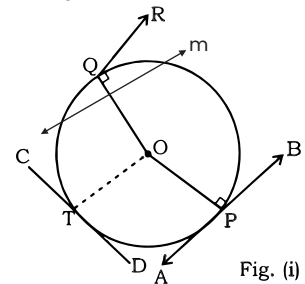


Fig. (i)

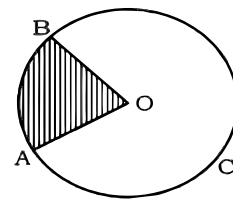


Fig. (ii)

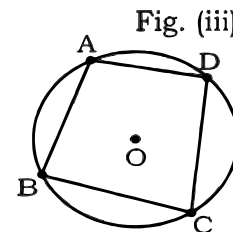


Fig. (iii)

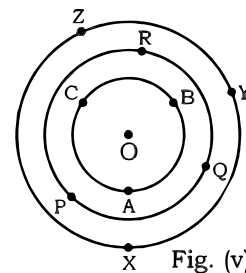


Fig. (v)

EXERCISE 7.2

1. Select 'False' statements out of the following statements.

- (i) A rectangle is a parallelogram.
- (ii) A square is a parallelogram as well as a rectangle.
- (iii) In a $\parallel$, interior angles on the same side of a transversal are supplementary.
- (iv) If a transversal intersects two lines in the same plane, then the alternate interior angles are not always congruent.
- (v) If the interior angles on the same side of a transversal are supplementary, the transversal intersects two lines in the same plane which do not intersect.
- (vi) If the diagonals of a quadrilateral bisect each other at right angles, the quadrilateral is a parallelogram.
- (vii) In a rectangle, the diagonals do not bisect each other at right angles.
- (viii) In a $\parallel$, if a diagonal is drawn, it makes two pairs of alternate angles, each pair consisting of congruent angles.
- (ix) A square is a regular quadrilateral.
- (x) If two lines never intersect however long they are produced, still they may or may not be parallel.

2. Perimeter of a $\parallel$ **PQRS** is 18 cm. If one side is twice that of the other, find the measure of each side.

3. In $\parallel$ **ABCD**, sum of measures of one pair of opposite angles is 130° , find the measure of each angle.

4. In $\parallel$ **WXYZ**, if $m\angle X$ is twice of $m\angle W$, find the measure of each angle.

5. In a $\parallel$ **ABCD**, diagonal **AC** is twice that of **BD**. If the sum of the measures of both the diagonals is 9 cm, find the measure of each diagonal. If diagonals intersect at point **P**, find **mPA** and **mPB** and express the relation **mPA : mPB**.

6. Define the following:

- | | |
|-----------------------|-------------------------|
| (i) Parallelogram | (vi) Sector |
| (ii) Regular Pentagon | (vii) Tangent |
| (iii) Regular Octagon | (viii) Concyclic points |
| (iv) Chord | (ix) Concentric circles |
| (v) Secant | (x) Diameter |

7. The radius of a circle whose centre is O is 3 cm long. Three points P, Q, R are in the same plane such that the distance of P from O is 2.5 cm, the distance of point Q from O is 4 cm and the distance of R from O is 3 cm. Discuss the location of each point with reference to the given circle.

8. Select 'True' statements out of the following given statements. Rewrite the statement correctly if it is 'False'.

- (i) A diameter passing through the centre of a circle is called a chord.
- (ii) A portion of circular region enclosed by two radii and one sector is called an arc.
- (iii) Secant is a line segment intersecting a circle at two points.
- (iv) A point lies in the exterior of a circle if its distance from the circle is greater than its radius.
- (v) Three or four points are said to be concyclic if a circle can pass through those points.
- (vi) Vertices of a $\parallel$ are concyclic.
- (vii) If four points are concyclic, opposite angles of the quadrilateral formed by these points are complementary.